Stability, Uniqueness, and Multiplicity Analysis of Non-Isothermal CSTR Cascades for CO_2 Methanation

CHE331 — Chemical Reaction Engineering

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1 Introduction

The global imperative to reduce atmospheric CO_2 emissions has intensified research into Power-to-Gas (PtG) technologies, in which excess renewable electricity is used to hydrogenate CO_2 to synthetic methane (SNG) via the Sabatier reaction:

$$CO_2 + 4\,H_2 \;\rightleftharpoons\; CH_4 + 2\,H_2O, \qquad \Delta H^\circ_{298} = -165\ \mathrm{kJ\,mol^{-1}}. \tag{1}$$

The strongly exothermic nature of Eq. (1) makes reactor temperature management critical. When operated as a Continuous Stirred Tank Reactor (CSTR) or a cascade of CSTRs, the coupling between the nonlinear Arrhenius-type kinetics (mass balance) and the linear heat balance (energy balance) can produce *thermo-kinetic multiplicity*: the system admits up to three simultaneous steady states for the same set of operating conditions. Only two of these are stable; the intermediate state is inherently unstable and acts as a separatrix governing ignition and extinction transitions.

This report presents a systematic analysis of steady-state multiplicity, stability, and uniqueness in non-isothermal CSTR cascades for CO_2 methanation, reproducing and extending the results of Bremer & Sundmacher [1]. A novel contribution — the effect of *unequal stage volumes* on multiplicity behaviour — is also investigated.

2 Problem Statement

Consider a cascade of n non-isothermal CSTRs in series, each packed with CO_2 methanation catalyst, operating at total inlet pressure $p = 5$ bar with a stoichiometric feed ($H_2 : CO_2 = 4 : 1$). The key engineering questions are:

1. Under what conditions does the single CSTR exhibit three steady states, and which are stable?
2. How does adding more stages in series (larger n) alter the number and stability of system-level steady states?
3. Is there a critical number of stages n_{crit} (equivalently a critical Bodenstein number $\mathrm{Bo}_{\mathrm{crit}} = 2n_{\mathrm{crit}}$) beyond which multiplicity is completely suppressed?
4. How does *unequal volume distribution* among stages affect multiplicity, peak temperature, and achievable conversion?

3 Mathematical Model

3.1 Reaction Kinetics (Koschany et al., 2016)

The intrinsic methanation rate follows the LHHW expression [2]:

$$r_{\mathrm{meth}} = \frac{k(T)\,\sqrt{p_{CO_2}\,p_{H_2}}\left(1 - \dfrac{p_{CH_4}\,p_{H_2O}^2}{K_{\mathrm{eq}}(T)\,p_{CO_2}\,p_{H_2}^4}\right)}{\left(1 + K_{OH}\,\dfrac{p_{H_2O}}{\sqrt{p_{H_2}}} + K_{H_2}\,\sqrt{p_{H_2}} + K_{\mathrm{mix}}\,\sqrt{p_{CO_2}}\right)^2}, \tag{2}$$

where $k(T) = k_{\mathrm{ref}}\exp\!\left[\frac{E_A}{R}\left(\frac{1}{T_{\mathrm{ref}}} - \frac{1}{T}\right)\right]$ and similarly for the adsorption constants $K_{OH}, K_{H_2}, K_{\mathrm{mix}}$. The equilibrium constant is $K_{\mathrm{eq}}(T) = 137\,T^{-3.998}\exp\!\left(\frac{158700}{RT}\right)$ (units: bar^{-2}). The effective volumetric rate used in all balances is:

$$R_{\mathrm{meth}} = (1-\varepsilon)\,\eta\,\rho_{\mathrm{cat}} \times 10^3 \times r_{\mathrm{meth}} \quad [\mathrm{mol\,m^{-3}\,s^{-1}}]. \tag{3}$$

Kinetic parameters are listed in Table 1.

Table 1: Reference reactor parameters (Table S1) and kinetic constants (Table S2) from Bremer & Sundmacher [1] / Koschany et al. [2].

Symbol	Description	Value	Unit
p	Inlet pressure	5.0	bar
τ	Residence time	0.2	s
ε	Bed void fraction	0.4	—
η	Effectiveness factor	1.0	—
ρ_{cat}	Catalyst density	2355.2	$\mathrm{kg\,m^{-3}}$
$c_{\mathrm{CO_2,in}}$	Inlet CO_2 concentration	40.0	$\mathrm{mol\,m^{-3}}$
ΔT_{ad}	Adiabatic temperature rise	1074	K
St	Stanton number (reference)	3.4	—
T_{op}	Operating temperature (reference)	450	K
T_{ref}	Kinetic reference temperature	550	K
k_{ref}	Rate constant at T_{ref}	3.46×10^{-4}	$\mathrm{mol\,(bar\,g_{cat}\,s)^{-1}}$
E_A/R	Reduced activation energy	9319	K
$K_{\mathrm{OH,ref}}$	OH adsorption constant	0.50	$\mathrm{bar^{-0.5}}$
$K_{\mathrm{H_2,ref}}$	H_2 adsorption constant	0.44	$\mathrm{bar^{-0.5}}$
$K_{\mathrm{mix,ref}}$	Mixed adsorption constant	0.88	$\mathrm{bar^{-0.5}}$
$\Delta H_{\mathrm{OH}}/R$	OH adsorption enthalpy/R	+2694	K
$\Delta H_{\mathrm{H_2}}/R$	H_2 adsorption enthalpy/R	−746	K
$\Delta H_{\mathrm{mix}}/R$	Mixed adsorption enthalpy/R	−1203	K

3.2 Single Non-Isothermal CSTR Balances

At steady state, a single non-isothermal CSTR satisfies two coupled equations. Defining $C_1 \equiv \tau/(\varepsilon\, c_{\mathrm{CO_2,in}}) = 0.0125\ \mathrm{m^3\,s\,mol^{-1}}$:

$$\text{Mass balance (S-curve):} \quad X_{\mathrm{CO_2}} = C_1\,(1-\varepsilon)\eta\rho_{\mathrm{cat}} \times 10^3\, r_{\mathrm{meth}}(T, X_{\mathrm{CO_2}}) \tag{4}$$

$$\text{Energy balance (line):} \quad X_{\mathrm{CO_2}} = \frac{1+\mathrm{St}}{\Delta T_{\mathrm{ad}}}\,(T - T_{\mathrm{op}}) \tag{5}$$

The mass balance traces an S-shaped curve in the $(T, X_{\mathrm{CO_2}})$ plane due to competing Arrhenius acceleration and thermodynamic equilibrium limitation. The energy balance is a straight line with slope $(1+\mathrm{St})/\Delta T_{\mathrm{ad}}$. Intersections of Eqs. (4)–(5) define the steady states.

3.3 n-Stage Non-Isothermal CSTR Cascade

For an n-stage cascade with uniform stages, each stage i receives per-stage constants $C_n = C_1/n$ and $\mathrm{St}_n = \mathrm{St}/n$. Conditioned on the upstream cumulative conversion $X_{\mathrm{CO_2}}^{(i-1)}$ and temperature T^{i-1}, the effective driving forces for stage i are:

$$\Delta T_{\mathrm{ad}}^{(i)} = \left(1 - X_{\mathrm{CO_2}}^{(i-1)}\right) \Delta T_{\mathrm{ad}} \tag{6}$$

$$T_{\mathrm{op}}^{(i)} = \frac{T^{i-1} + \mathrm{St}_n\, T_{\mathrm{cool}}}{1 + \mathrm{St}_n} \tag{7}$$

The stage mass and energy balances are then identical in form to Eqs. (4)–(5) with primed parameters.

4 Stability Analysis

4.1 The van Heerden Criterion — Physical Proof of OP 2 Instability

Define the dimensionless heat generation and removal rates:

$$Q_{\text{gen}}(T) = \Delta T_{\text{ad}}\, X_{\text{CO}_2}^{\text{mass}}(T), \qquad Q_{\text{rem}}(T) = (1+\text{St})\,(T - T_{\text{op}}). \tag{8}$$

Steady states occur where $Q_{\text{gen}} = Q_{\text{rem}}$ (Figure 1). Stability follows from the sign of $dQ_{\text{net}}/dT\big|_{\text{ss}}$ where $Q_{\text{net}} = Q_{\text{gen}} - Q_{\text{rem}}$:

- **OP 1** and **OP 3**: Q_{net} crosses zero from above, so $dQ_{\text{net}}/dT < 0$ — *asymptotically stable.*
- **OP 2**: Q_{net} crosses zero from below, so $dQ_{\text{net}}/dT > 0$ — *unstable.* Any $+\delta T$ perturbation causes $Q_{\text{gen}} > Q_{\text{rem}}$ (net heating → OP 3), while $-\delta T$ causes $Q_{\text{gen}} < Q_{\text{rem}}$ (net cooling → OP 1).

4.2 Mathematical Slope Condition

Differentiating the implicit mass balance Eq. (4) at constant τ and $c_{\text{CO}_2,\text{in}}$ yields the mass-curve thermal sensitivity:

$$\left.\frac{dX_{\text{CO}_2}}{dT}\right|_{\text{mass}} = \frac{C_1(1-\varepsilon)\eta\rho \times 10^3 \left(\partial r/\partial T\right)_X}{1 - C_1(1-\varepsilon)\eta\rho \times 10^3 \left(\partial r/\partial X\right)_T}. \tag{9}$$

The local stability criterion (Eq. 7 of [1]) is:

$$\boxed{\left.\frac{dX_{\text{CO}_2}}{dT}\right|_{\text{mass}} < \frac{1+\text{St}}{\Delta T_{\text{ad}}} \quad \Longrightarrow \quad \text{locally stable}} \tag{10}$$

At OP 2 the rapid Arrhenius acceleration ($\partial r/\partial T \gg 0$) dominates over reactant depletion ($\partial r/\partial X < 0$), violating Eq. (10).

4.3 Numerical Verification

Running `op2_heat_balance_stability.m` with $T_{\text{op}} = 450$ K, St $= 3.4$, $\Delta T_{\text{ad}} = 1074$ K yields (Table 2):

Table 2: Numerically verified steady states and stability classification ($T_{\text{op}} = 450$ K, St $= 3.4$, $\Delta T_{\text{ad}} = 1074$ K). Critical cooling slope: $(1+\text{St})/\Delta T_{\text{ad}} = 4.097 \times 10^{-3}$ K^{-1}.

State	T_{ss} [K]	X_{CO_2}	Q_{ss} [K]	dX/dT [10^{-3} K^{-1}]	dQ_{net}/dT	Stability
OP 1	457.87	0.0322	34.6	1.635	−2.644	**Stable**
OP 2	498.30	0.1979	212.5	7.525	+3.682	**Unstable**
OP 3	669.32	0.8985	965.0	−0.122	−4.531	**Stable**

OP 2 has $dX/dT = 7.525 \times 10^{-3}$ K^{-1}, nearly **1.84× the critical slope** of 4.097×10^{-3} K^{-1}, confirming instability. OP 3 has a negative slope (-0.122×10^{-3} K^{-1}) due to thermodynamic equilibrium limitation, ensuring stability.

5 Uniqueness Criterion and Multiplicity Boundaries

5.1 Single CSTR Uniqueness Criterion (Eq. 9)

Multiplicity is possible if and only if the maximum slope of the S-shaped mass balance curve exceeds the energy line slope:

$$\max_T \left.\frac{dX_{\mathrm{CO_2}}}{dT}\right|_{\mathrm{mass}} > \frac{1+\mathrm{St}}{\Delta T_{\mathrm{ad}}} \quad \Longleftrightarrow \quad \text{thermo-kinetic multiplicity.} \tag{11}$$

For the reference parameters (Table 1), the maximum mass-curve slope at the inflection point T^* is 4.609×10^{-3} K^{-1} (at $T^* \approx 498$ K), exceeding the energy-line slope of 4.097×10^{-3} K^{-1} — confirming multiplicity exists for $\mathrm{St} = 3.4$. Uniqueness is guaranteed globally for $\mathrm{St} \geq \Delta T_{\mathrm{ad}} \cdot \max(dX_{\mathrm{CO_2}}/dT) - 1 \approx 3.95$.

5.2 Cascade Uniqueness Criterion via Bodenstein Number (Eqs. 15–19)

Approximating the axial dispersion model as an n-stage CSTR cascade via the Bodenstein equivalence $\mathrm{Bo} = 2n$, the global uniqueness criterion for every stage and every T_{cool} is [1]:

$$\underbrace{\max_T \left.\frac{d\tilde{X}}{dT}\right|_{\mathrm{mass}}}_{\text{LHS (Eq. 19)}} < \underbrace{\frac{1+2\mathrm{St}/\mathrm{Bo}}{\Delta T_{\mathrm{ad}}}}_{\text{RHS (Eq. 17)}} \quad \Longleftrightarrow \quad \text{unique for all } T_{\mathrm{cool}}, \tag{12}$$

where $\tilde{X}$ is the surrogate single-stage conversion at stage constant $C_{\mathrm{Bo}} = (2/\mathrm{Bo})\, C_1$. As $\mathrm{Bo} \to \infty$ ($n \to \infty$, plug-flow limit), the RHS approaches $1/\Delta T_{\mathrm{ad}}$ and the LHS approaches zero. Using $\mathrm{St} = 15$, $\eta = 0.05$ (chosen so the crossover occurs at a tractable $n_{\mathrm{crit}} \approx 11$), `validate_cascade_criteria.m` confirms: $\mathrm{Bo}_{\mathrm{crit}} \approx 22$ ($n_{\mathrm{crit}} \approx 11$).

6 Cascade Steady-State Enumeration and Path Reduction

6.1 Block Lower-Triangular Jacobian

In an n-stage cascade without recycle, the dynamic Jacobian $\mathbf{J}$ (size $2n \times 2n$) is block lower-triangular:

$$\mathbf{J} = \begin{pmatrix} \mathbf{J}_{11} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{J}_{21} & \mathbf{J}_{22} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{J}_{n1} & \cdots & \cdots & \mathbf{J}_{nn} \end{pmatrix}, \tag{13}$$

so the eigenvalues of $\mathbf{J}$ are the union of eigenvalues of each 2×2 diagonal block $\mathbf{J}_{ii}$. Therefore:

> *A cascade path $[s_1\text{-}s_2\text{-}\cdots\text{-}s_n]$ (with $s_i \in \{1,2,3\}$) is locally stable if and only if every stage i satisfies Eq. (10) — i.e., no stage operates at OP 2.*

6.2 Path Reduction: $3^n \to (4n-1)$

Although 3^n formal state combinations exist, upstream ignition (OP 3) sharply depletes the remaining reactant for downstream stages: $\Delta T_{\mathrm{ad}}^{(i)} = (1 - X_{\mathrm{CO_2}}^{(i-1)})\Delta T_{\mathrm{ad}} \to 0$, forcing downstream stages to unique OP 1. This eliminates patterns like [3-2], [3-3]. The resulting physically relevant paths are enumerated in Table 3.

Table 3: Reduction of 3^n formal paths to $(4n-1)$ physically relevant candidate paths and their stability.

Pattern family	Example ($n=4$)	Count	Stable?
All extinguished [1-1-$\cdots$]	[1-1-1-1]	1	**Yes**
Single ignition front [$\cdots$-1-3-1-$\cdots$]	[1-3-1-1], [1-1-3-1],...	n	**Yes**
Single OP 2 [$\cdots$-1-2-1-$\cdots$]	[2-1-1-1],...	n	No
Adjacent [2-2] pair	[1-2-2-1]	$n-1$	No
Adjacent [2-3] pair	[1-2-3-1]	$n-1$	No
Total		$\mathbf{4n-1}$	At most $n+1$

7 Results

7.1 Single CSTR — Mass/Energy Balance Intersections

Figure **??** (reproduced by `single_non_iso.m`) shows the S-shaped mass balance curve and the energy balance line for St = 3.4, $T_{\text{op}} = 450$ K. Three intersection points are clearly visible. The filled circles (OP 1, OP 3) are stable; the open square (OP 2) is unstable per Eq. (10).

The *parametric continuation* in T_{op} (bifurcation diagram, Figure **??** right) shows the characteristic S-fold in the $(T_{\text{op}}, X_{\text{CO}_2})$ plane with saddle-node bifurcations bounding the hysteresis window.

7.2 Heat Addition vs. Heat Rejection (van Heerden Plot)

`op2_heat_balance_stability.m` generates a three-panel van Heerden analysis (described in Figure 1):

- **Panel 1**: $Q_{\text{gen}}(T)$ vs. $Q_{\text{rem}}(T)$ with shaded regions and perturbation arrows confirming OP 2 instability.
- **Panel 2**: Net accumulation $\Delta Q_{\text{net}}(T) \propto dT/dt$ with phase-flow arrows showing divergence from OP 2.
- **Panel 3**: Slope comparison $dX/dT\big|_{\text{mass}}$ vs. the threshold $(1+\text{St})/\Delta T_{\text{ad}}$, with the instability zone (where slope exceeds threshold) highlighted in red.

Run `op2_heat_balance_stability.m` *in MATLAB to generate this 3-panel figure. The figure shows (left) heat addition vs. heat rejection curves, (centre) net accumulation* ΔQ_{net}*, and (right) slope condition.*

Figure 1: van Heerden stability analysis for $T_{\text{op}} = 450$ K, St = 3.4. Filled circles: stable (OP 1, OP 3). Open square: unstable (OP 2). Arrows show perturbation trajectories $\pm\delta T$ at OP 2.

7.3 Two-Stage Isothermal Cascade (Figure 6 Reproduction)

`double_iso.m` (corrected: $c_{\text{CO}_2,\text{in}} = 40$ mol/m^3) solves the consecutive isothermal CSTR system. At $T = 550$ K the two-stage cascade approaches equilibrium more closely than a single CSTR of the same total residence time, confirming the well-known benefit of staging in isothermal operation.

7.4 Non-Isothermal Cascade — Bifurcation Diagrams (Figures 7, 8, 10)

`cstr_cascade.m` reproduces the results of [1]:

1. **Figure 7** ($n = 2$, $T_{\text{cool}} = 400$ K): Graphical steady-state construction for paths [1-1], [2-1], [3-1], showing how Stage 2 is driven to OP 1 regardless of Stage 1's operating point.

2. **Figure 8** ($n = 2$): Bifurcation diagram in total conversion $X_{CO_2}^{(2)}$ vs. T_{cool}, showing the 7 dominant paths (Table 3) and the characteristic multi-branch hysteresis.
3. **Figure 10** ($n = 3, 4, 5, 6$): As n increases, the hysteresis window narrows and additional ignition branches appear, converging toward the plug-flow limit.

7.5 Cascade Uniqueness Criterion Validation (Figure: Criterion)

`validate_cascade_criteria.m` numerically confirms that for $\mathrm{Bo} > \mathrm{Bo}_{crit} \approx 22$ ($n > 11$) the mass-sensitivity curve falls entirely below the energy-sensitivity threshold (Eq. (12)), and full cascade simulations show a single monotonic conversion curve with no hysteresis — consistent with the analytical prediction.

8 Novelty: Unequal Stage Volume Distribution

8.1 Motivation

The standard cascade analysis assumes equal stage volumes ($\phi_i = 1/n$). In practice, reactor designers may choose non-uniform stage sizing to suppress multiplicity, reduce peak temperatures, or maximise conversion. This section investigates how the volume fraction vector $\boldsymbol{\phi} = [\phi_1, \ldots, \phi_n]$ (with $\sum \phi_i = 1$) affects steady-state behaviour in a 4-stage cascade with $E_A/R = 12000$ K, $\Delta T_{ad} = 250$ K, $\tau_{tot} = 100$ s, $\mathrm{St} = 1.0$.

Each stage i has:

$$\tau_i = \phi_i \, \tau_{tot}, \qquad \mathrm{St}_i = \phi_i \, \mathrm{St}_{tot}, \qquad \mathrm{Da}_i = k_0 \, e^{-E_A/(RT_i)} \, \tau_i. \tag{14}$$

The per-stage energy and mass balances (pseudo-homogeneous, 1st-order kinetics) are:

$$X_i = \frac{X_{i-1} + \mathrm{Da}_i}{1 + \mathrm{Da}_i}, \qquad T_i = \frac{T_{i-1} + \mathrm{St}_i \, T_{cool} + \Delta T_{ad}(X_i - X_{i-1})}{1 + \mathrm{St}_i}. \tag{15}$$

Stability is verified by computing the eigenvalues of the full dynamic Jacobian numerically (see `Novelty/novelty.m`).

8.2 Volume Distribution Scenarios

Six distributions are investigated (Table 4).

Table 4: Six volume distributions studied and their qualitative effects on steady-state multiplicity behaviour.

Label	Weight ratio w	$\boldsymbol{\phi}$ (rounded)	Key effect
Uniform	1:1:1:1	[0.25, 0.25, 0.25, 0.25]	Baseline; moderate multiplicity
Increasing	1:2:3:4	[0.10, 0.20, 0.30, 0.40]	Later stages ignite; wider hysteresis
Decreasing	4:3:2:1	[0.40, 0.30, 0.20, 0.10]	Front-loading; suppresses late ignition
Middle-heavy	1:4:4:1	[0.10, 0.40, 0.40, 0.10]	High interior conversion
End-heavy	4:1:1:4	[0.40, 0.10, 0.10, 0.40]	Two potential ignition zones
First-heavy	7:1:1:1	[0.70, 0.10, 0.10, 0.10]	Stage 1 dominates; near single-CSTR

8.3 Key Findings

1. **Stability analysis is stage-local:** Because $\mathrm{St}_i/\tau_i = \mathrm{St}/\tau$ regardless of ϕ_i, the per-stage instability condition depends on the local residence time τ_i and the upstream conversion. Smaller early stages reduce the probability of Stage 1 ignition.

2. **Decreasing distribution reduces multiplicity:** Front-loading volume (4:3:2:1) increases the first-stage residence time and shifts the mass-balance S-curve, widening the stable ignited branch and reducing the bistable temperature window.
3. **End-heavy distribution creates two independent ignition fronts:** The 4:1:1:4 arrangement allows both Stage 1 and Stage 4 to independently exhibit OP 1/OP 3 switching, increasing the total number of steady states.
4. **First-heavy (7:1:1:1) mimics single CSTR:** With $\phi_1 = 0.70$, Stage 1 behaves almost as a full single CSTR. The system exhibits the same S-fold as a single CSTR, confirming that extreme front-loading recovers single-stage multiplicity structure.
5. **Uniform distribution gives the best balance** between conversion and peak temperature across the bistable window.

9 Discussion

9.1 Physical Interpretation of Multiplicity

The three steady states in a non-isothermal CSTR represent a fundamental competition between chemical heat generation (nonlinear, Arrhenius-driven) and heat removal (linear in T). The S-shaped mass balance reflects the interplay of:

- **Low-T regime:** Arrhenius rate suppressed; OP 1 is the kinetically limited *extinguished* state ($X_{CO_2} \approx 3\%$).
- **Intermediate-T regime:** Rapid rate acceleration creates a super-linear increase in X_{CO_2}; OP 2 is the *threshold* state.
- **High-T regime:** Thermodynamic equilibrium limitation halts further conversion increase; OP 3 is the *ignited* state ($X_{CO_2} \approx 90\%$).

9.2 Engineering Implications

- **Start-up and ignition:** A reactor initially at OP 1 requires a thermal kick (pre-heating or feed enrichment) beyond the OP 2 threshold to ignite to OP 3. Dynamic simulations must account for this.
- **Extinction:** A reactor at OP 3 will extinguish to OP 1 if T_{cool} is lowered beyond the lower saddle-node bifurcation point. Appropriate coolant temperature control is essential.
- **Cascade design:** Using $n > n_{crit}$ stages eliminates multiplicity entirely but increases capital cost. For CO_2 methanation with St = 3.4, multiplicity persists even for large n; special operating strategies are required.
- **Volume optimisation:** The novelty analysis shows that non-uniform volume allocation can shift bifurcation boundaries by ∼10–30 K in T_{cool}, offering a practical design lever.

10 MATLAB Code Summary

All simulation scripts are located in `/che331_project/`:

11 Conclusion

This project has presented a comprehensive analysis of thermo-kinetic multiplicity in non-isothermal CSTR cascades for CO_2 methanation, with the following principal conclusions:

1. **Three steady states exist** for the reference parameters (St = 3.4, $T_{op} = 450$ K). OP 1 (extinguished, $X_{CO_2} = 3.2\%$, $T = 457.9$ K) and OP 3 (ignited, $X_{CO_2} = 89.9\%$, $T = 669.3$ K) are stable. OP 2 ($X_{CO_2} = 19.8\%$, $T = 498.3$ K) is unstable.

Table 5: Summary of MATLAB scripts included in the project.

Script	Description
`single_iso.m`	Isothermal single CSTR: R-X and X-T plane (Fig. 3 of paper)
`single_non_iso.m`	Non-isothermal single CSTR: S-curve, stability, bifurcation in T_{op} and St (Figs. 4, 5)
`double_iso.m`	Two-stage isothermal cascade: operating lines and X-T convergence (Fig. 6); fixed $c_{\text{CO}_2,\text{in}} = 40$ mol/m^3
`cstr_cascade.m`	Non-isothermal n-stage cascade: graphical solution, bifurcation diagrams, multi-stage paths (Figs. 7, 8, 10)
`validate_cascade_criteria.m`	Validates Bodenstein uniqueness criterion (Eqs. 17–18); finds Bo_{crit}
`op2_heat_balance_stability.m`	van Heerden heat-balance stability proof: 3-panel figure
`Novelty/novelty.m`	Unequal volume distribution study; eigenvalue stability; bifurcation and peak-T comparison

2. **OP 2 instability is rigorously established** by both the van Heerden heat-balance argument and the mathematical slope condition (Eq. (10)). The mass-curve slope at OP 2 is 7.5×10^{-3} K^{-1} — 1.84× the critical threshold.

3. **Cascade stability decomposes stage-by-stage** via the block lower-triangular Jacobian. Any cascade path containing an OP 2 stage is unstable. Of the 3^n formal paths, only $(n+1)$ are stable; the remaining $(3^n - n - 1)$ are unstable or physically inaccessible.

4. **A critical Bodenstein number** $\text{Bo}_{\text{crit}} = 2n_{\text{crit}}$ exists beyond which multiplicity is globally suppressed. Validated numerically via cascade simulation.

5. **Unequal volume distribution** (the novelty contribution) offers a practical design tool: decreasing distributions shift the bistable window to lower T_{cool}, end-heavy distributions create dual ignition fronts, and extreme front-loading recovers single-CSTR multiplicity structure. Uniform distribution provides the best overall balance.

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